PHYS 350: Honours Electricity and Magnetism

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Abstract

NOT FINISHED

Contents

1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Taught by Debanjan Sarkar & Adrian C. Liu. Mondays, Wednesdays and Fridays from 13:35 to 14:25 in rutherford 112. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

2.1 Scalars, Vectors, and Fields

Definition 1 (Scalar)**.** *A scalar is a quantity with just a magnitude*

Definition 2 (Vector)**.** *Vector is a quantity with a magnitude and a direction. A vector can be written in the form of*

$$\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}.$$

Definition 3 (Cross Product)**.** *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors, and can be mathematically written as*

$$\vec{v} \times \vec{w} = |\vec{v}||\vec{w}| \sin\theta\, \hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}.$$

Definition 4 (Dot Product)**.** *The dot product measures how much of $\vec{v}$ points along $\vec{w}$, so it is zero for perpendicular vectors, and can be mathematically written as*

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z = |\vec{v}||\vec{w}| \cos\theta.$$

Remark 1. *Note $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, and $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$.*

Definition 5 (Field)**.** *A field is something with a value at every point in space. A scalar field assigns a number to each point, like temperature $T(x, y, z)$ or the height of a hill. A vector field assigns a vector to each point, like fluid velocity or $\vec{E}$.*

2.2 Divergence, Gradient, and Curl

Define the operator

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right),$$

called del. It means nothing until it is fed a field. Then

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z} \qquad \text{(gradient: scalar} \to \text{vector),} \tag{1}$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \qquad \text{(divergence: vector} \to \text{scalar),} \tag{2}$$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \qquad \text{(curl: vector} \to \text{vector).} \tag{3}$$

For the divergence we differentiate each component along its own direction. For the curl we mix directions, which is why terms like $\partial F_y/\partial x$ show up. The direction of the curl comes from the right hand rule: curl your fingers the way the paddle wheel turns and your thumb points along $\vec{\nabla} \times \vec{F}$.

The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f.

2.3 Integrals

An integral is an instruction to move along a region, evaluate something at each step, and add it to a running total. The multivariable versions are:

$$\int_C \vec{F} \cdot d\vec{l} \qquad \text{line integral, the work done if } \vec{F} \text{ is a force,} \tag{4}$$

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n}\, da \qquad \text{flux, how much of the field pokes through the surface,} \tag{5}$$

$$\int_V f\, d^3\vec{r} \qquad \text{volume integral, the mass if } f \text{ is a density.} \tag{6}$$

We write $\oint$ when the curve is a closed loop or the surface is closed.

Example. *Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R. On the circle $d\vec{l} = R\, d\phi(-\sin\phi\, \hat{x} + \cos\phi\, \hat{y})$ and $y = R\sin\phi$, so*

$$\int_C \vec{F} \cdot d\vec{l} = -R^2 \int_0^{\pi/2} \sin^2\phi\, d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ points along $+\hat{x}$ while we travel in the $-\hat{x}$ direction over most of the arc.

2.4 The Three Fundamental Theorems

In single variable calculus, $\int_a^b (\partial f/\partial x)\,dx = f(b) - f(a)$. The same pattern holds in higher dimensions: integrating a derivative over a region reduces to evaluating the original object on the boundary of that region. The three theorems differ only in the dimension of the region.

Theorem 1 (Gradient theorem)**.** *The boundary of a curve is two points:*

$$\int_a^b \vec{\nabla} h \cdot d\vec{l} = h(b) - h(a).$$

To find my altitude change climbing a hill, I can compare before and after, or add up all the little changes along the way.

Theorem 2 (Stokes' theorem)**.** *The boundary of a surface is a curve:*

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_C \vec{F} \cdot d\vec{l}.$$

To get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside.

Theorem 3 (Gauss' divergence theorem)**.** *The boundary of a volume is a surface:*

$$\int_V (\vec{\nabla} \cdot \vec{F})\, d^3\vec{r} = \oint_S \vec{F} \cdot d\vec{a}.$$

To find the total flux leaking out through the surface, I can add up all the sources inside.

Remark 2. *Maxwell's equations come in a differential form written with $\vec{\nabla}\cdot$ and $\vec{\nabla}\times$, and an equivalent integral form written with $\oint \vec{F} \cdot d\vec{a}$ and $\oint \vec{F} \cdot d\vec{l}$. These two theorems are what let us pass between them. Gauss' law and Ampère's law are the first examples.*

2.5 Second Derivatives

Since we have three kinds of first derivative, we can mix and match to build second derivatives. Taking the divergence of a gradient gives

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f,$$

where ∇^2 is the Laplacian. Mathematicians often write Δ, but in physics we use ∇^2. Acting on a scalar field it returns a scalar field. It can also act on a vector field, component by component, returning a vector field.

The full list of second derivative identities:

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f \qquad \text{(generally non-zero)}, \tag{7}$$
$$\vec{\nabla} \times (\vec{\nabla} f) = 0 \qquad \text{(curl of a gradient)}, \tag{8}$$
$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0 \qquad \text{(divergence of a curl)}, \tag{9}$$
$$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F} \qquad \text{(the BAC–CAB of vector calculus)}. \tag{10}$$

These can be proved by writing out every partial derivative, which is tedious, or with index notation.

2.6 Index Notation

Write the ith component of $\vec{v}$ as v_i, so $v_1 = v_x$, $v_2 = v_y$, $v_3 = v_z$. A matrix carries two indices, A_{ij}, where i says which row and j says which column. A symmetric matrix has $A_{ij} = A_{ji}$, and an antisymmetric one has $B_{ij} = -B_{ji}$.

Definition 6 (Kronecker delta)**.** *The identity matrix is written*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Matrix multiplication $\vec{w} = A\vec{v}$ becomes $w_i = \sum_j A_{ij} v_j$. Multiplying by the identity gives $\sum_j \delta_{ij} v_j = v_i$, since the delta kills every term except $j = i$. This is the main trick: a delta summed against something sets one index equal to the other.

The dot product is $\vec{v} \cdot \vec{w} = \sum_i v_i w_i$, with no free index because the result is a scalar. The cross product needs a second symbol:

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k.$$

Definition 7 (Levi-Civita symbol)**.** *ε_{ijk} is defined by*

- $\varepsilon_{123} = +1$*, and cyclic permutations keep the sign:* $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312}$.
- *Swapping any two indices flips the sign, so* $\varepsilon_{213} = \varepsilon_{132} = -1$.
- *If any two indices are equal,* $\varepsilon_{ijk} = 0$.

The identity that does all the work is

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}, \tag{11}$$

which can be checked component by component.

Writing $\partial_i \equiv \partial/\partial x_i$, the derivatives become

$$\vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i, \qquad (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

2.7 Proving the Identities

Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \left(\sum_{m,n} \varepsilon_{kmn} \partial_m F_n \right) \tag{12}$$

$$= \sum_{j,m,n} \left(\sum_{k} \varepsilon_{kij} \varepsilon_{kmn} \right) \partial_j \partial_m F_n \tag{13}$$

$$= \sum_{j,m,n} \left(\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm} \right) \partial_j \partial_m F_n. \tag{14}$$

In the second line we used the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ to move the summed index to the front so the identity applies. The first term gives

$$\sum_j \partial_j \partial_i F_j = \partial_i \left(\sum_j \partial_j F_j \right) = \left(\vec{\nabla}(\vec{\nabla} \cdot \vec{F}) \right)_i ,$$

and the second gives $-\sum_j \partial_j \partial_j F_i = -(\nabla^2 \vec{F})_i$. Together they reproduce the identity claimed above.

For the curl of a gradient,

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f. \tag{$*$}$$

Now j and k are internal to the sum, so their names do not matter. Swap the labels:

$$= \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f = -\sum_{j,k} \varepsilon_{ijk} \partial_k \partial_j f = -\sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f. \tag{$**$}$$

The second step uses the antisymmetry of ε and the third uses the fact that partial derivatives commute. Comparing $(*)$ and $(**)$, the expression equals minus itself, and the only such number is zero. So $\vec{\nabla} \times (\vec{\nabla} f) = 0$.

2.8 Irrotational and Solenoidal Fields

The two vanishing identities let us classify fields.

- Since $\vec{\nabla} \times (\vec{\nabla}\psi) = 0$, writing $\vec{F} = \vec{\nabla}\psi$ guarantees $\vec{\nabla} \times \vec{F} = 0$. Such a field is **irrotational**.

- Since $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$, writing $\vec{F} = \vec{\nabla} \times \vec{A}$ guarantees $\vec{\nabla} \cdot \vec{F} = 0$. Such a field is **solenoidal**.

Any vector field can be written as $\vec{\nabla}\psi + \vec{\nabla} \times \vec{A}$.

Theorem 4 (Irrotational fields)**.** *The following four statements are equivalent:*

(i) $\vec{\nabla} \times \vec{F} = 0$ *everywhere.*

(ii) $\int_a^b \vec{F} \cdot d\vec{l}$ *is independent of path.*

(iii) $\oint_C \vec{F} \cdot d\vec{l} = 0$ *for any closed loop.*

(iv) $\vec{F} = \vec{\nabla}\psi$ *for some scalar* ψ*.*

Theorem 5 (Solenoidal fields)**.** *The following four statements are equivalent:*

(i) $\vec{\nabla} \cdot \vec{F} = 0$ *everywhere.*

(ii) $\int_S \vec{F} \cdot d\vec{a}$ *is independent of the surface, for surfaces sharing a boundary.*

(iii) $\oint_S \vec{F} \cdot d\vec{a} = 0$ *for any closed surface.*

(iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ *for some vector field* $\vec{A}$*.*

Remark 3. *In electrostatics we will find* $\vec{\nabla} \times \vec{E} = 0$*. That is the irrotational case, and it is what licenses us to write* $\vec{E} = -\vec{\nabla}V$ *and introduce the electric potential. In magnetostatics we will find* $\vec{\nabla} \cdot \vec{B} = 0$*, which is the solenoidal case and licenses* $\vec{B} = \vec{\nabla} \times \vec{A}$*, the vector potential.*

3 Electrostatics

4 Potentials

5 Electric Fields in Matter

6 Magnetostatics

7 Magnetic Fields in Matter

8 Electrodynamics

A Appendix

B Useful Links